TURNING WITH EASE

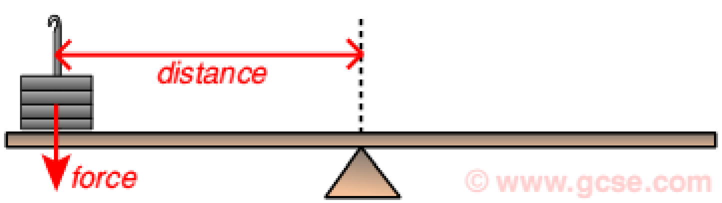
CHAPTER 4

Calculating moments

- Moment of a force = force x perpendicular distance from pivot to the force
- Moment= force (f) x perpendicular distance (d)

Moment = fd

- The units for moments is Newton meter (Nm)



Worked examples

Suppose the force applied in figure above is 40N and that the perpendicular distance is 2.0 m from the turning point. calculate the moment of the force

Solution

Moment=fd

Moment= 40n x 2.0m = 80 NM

Cont ‘d

Note;

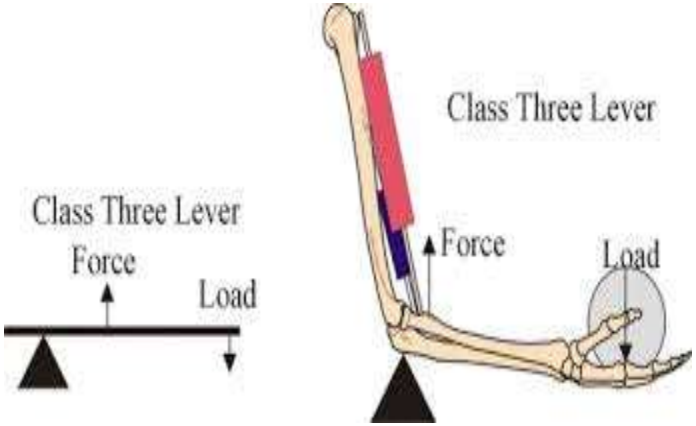
The method of application of a force depends on circular motion and the idea of proportion.

LEVER- is the simplest machine

- It is used to apply force to a load
- It is a simple rigid rod which can rotate freely around a pivot.
- Examples of levers in human body, forearm and leg

Human body lever

- Human forearm



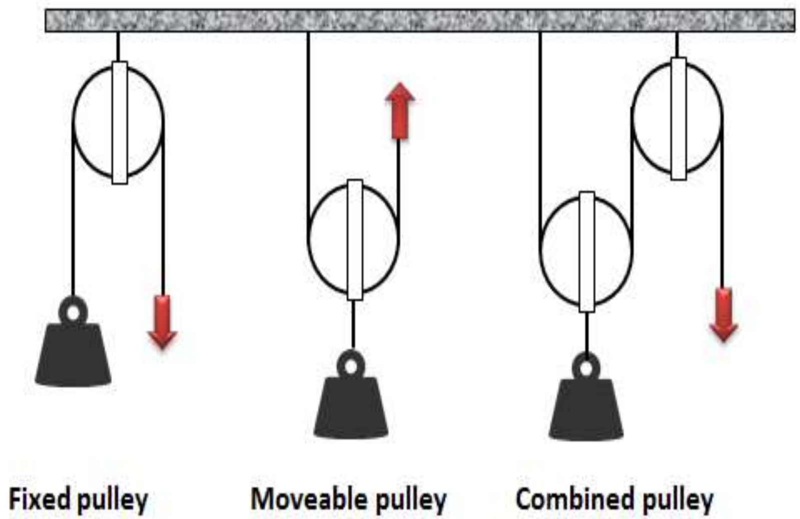
Pulley- is a flat circular disc usually made of metal or wood with a groove in its rim and a rope passing through the groove. It rotates at a fixed point called axle

Wheel-and -axle

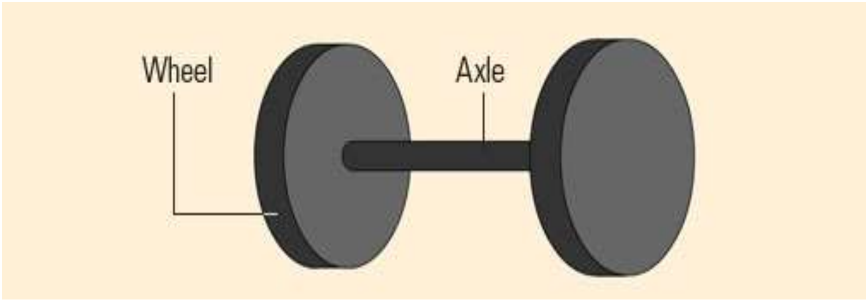
- This simple machine involves two circular objects — a larger disc and a smaller cylinder, both joined at the centre. The larger disc is called the wheel, and the smaller cylindrical object or rod is referred to as the axle.
- Sometimes, there may be two wheels attached to both ends of the axle. A wheel alone or an axle alone is not a simple machine. They need to be joined to be called a simple machine.
- The wheel is said to be the single most important invention of mankind

Pulley, wheel and axle

a) Pulley



b) wheel and axle



c) Others includes inclined plane, wedges screw among others

Equilibrium

- Equilibrium is the state of balance
- A body is said to be in equilibrium if;

1. The forces acting on it are balanced
2. The turning effect of the force on it is balanced

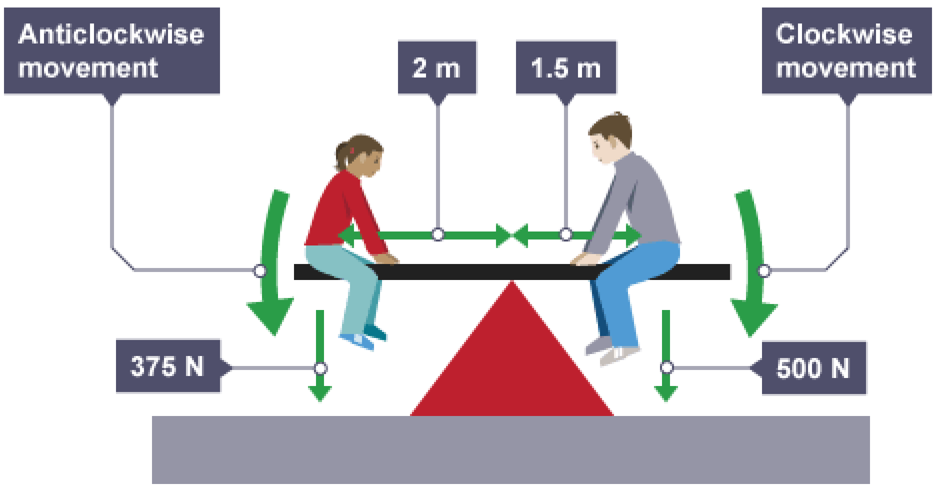
Note.

- If a resultant force acts on an object, it will start to move off in the direction of the resultant force
- If there is a resultant turning effect, it will start to rotate

Other examples of equilibrium; tug-of-war

Balancing moments

Beam balance

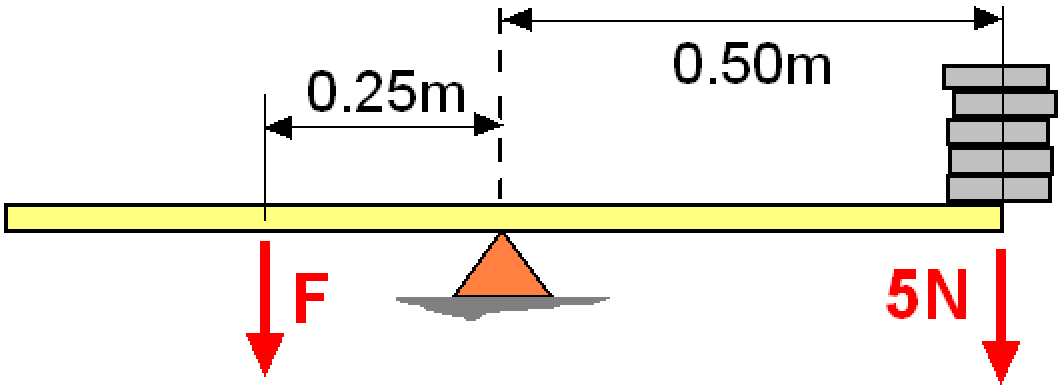


Law of moments

Total clockwise moment= total anticlockwise moment

Worked examples

Calculate the value of F

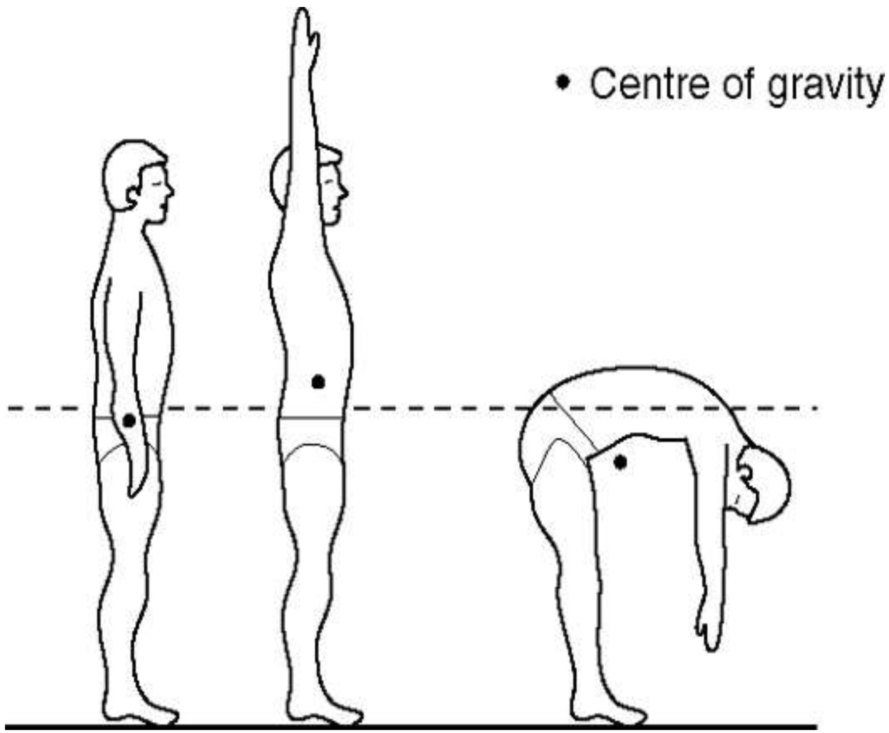


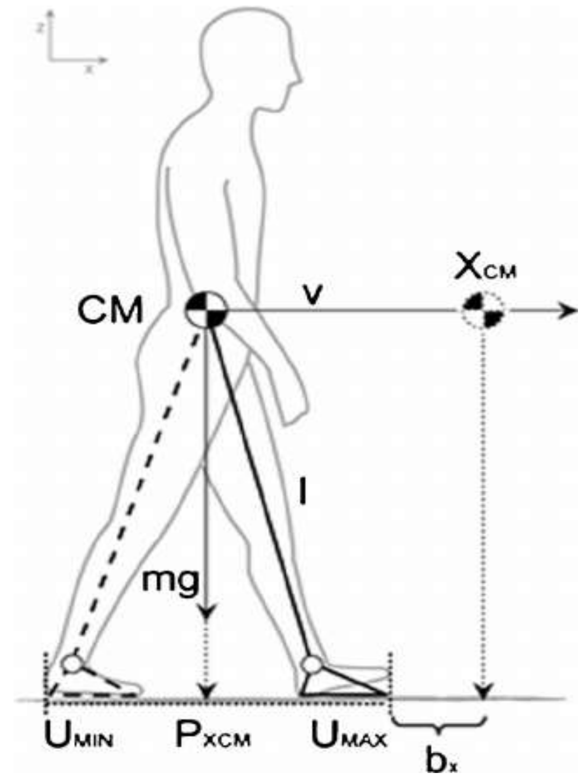
Stability and centre of mass

- Allah Ta’ala has created humans fairly symmetrical so their centre mass must lie somewhere on the axis of symmetry.
- One half is on one side of the axis and the other half on the other
- Centre of mass is the point at which the body experiences the total fall

Upright posture of humans

The centre of mass of a body is in the middle of the body roughly level with the navel





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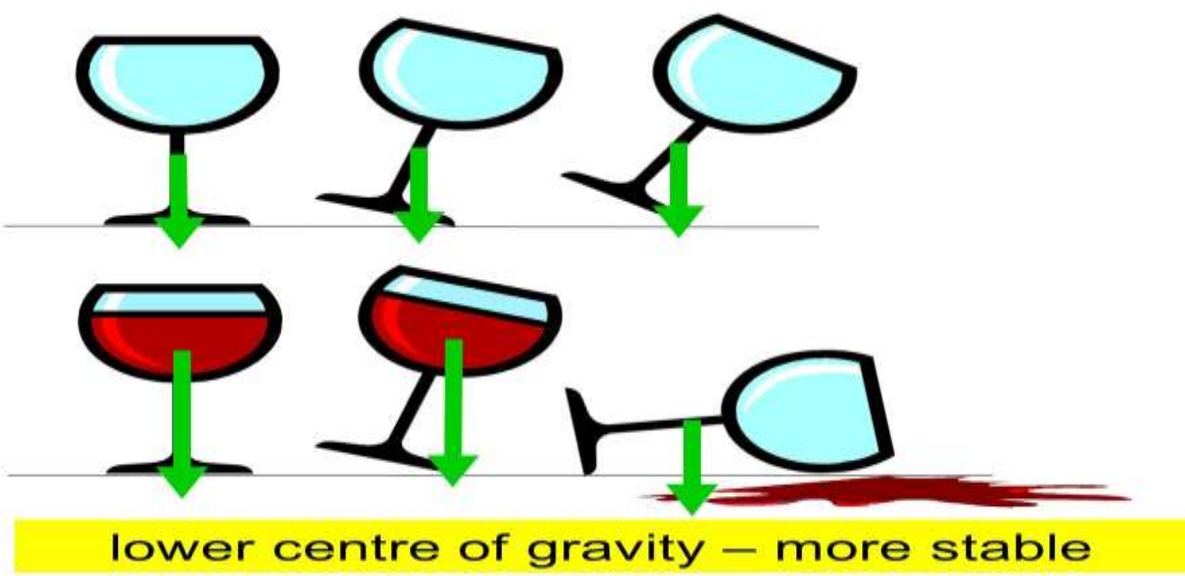
- Humans are inherently unstable
- It takes a lot of time for children learn to stand and walk.
- We have to learn how to coordinate our muscles so that our legs, body and arms move correctly
- This combination of muscles and organs working together highlights the importance of walking straight. Allah Ta’ala states

Is he who goes falling on his face more rightly guided or he who walks upright on a straight path.

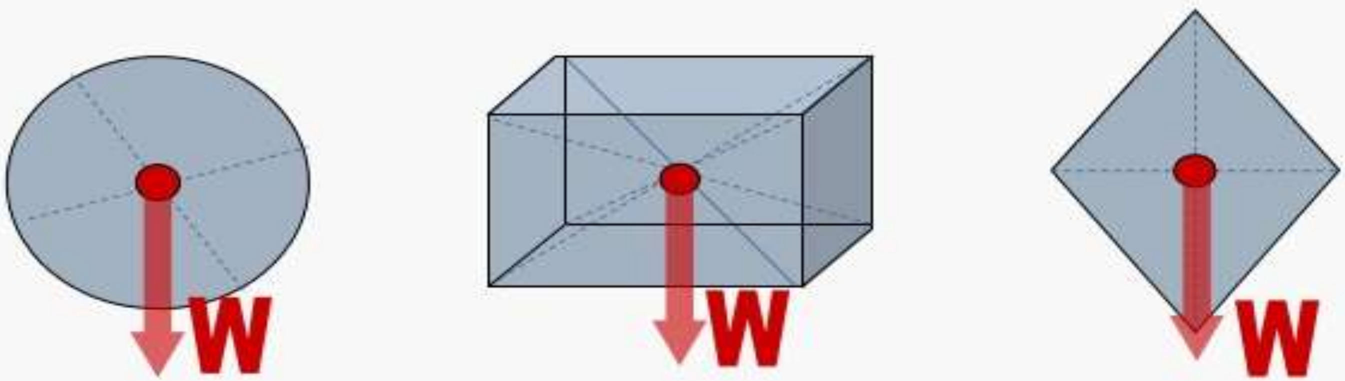
- A tall glass is easily knocked over- it is unstable
- When the glass is upright , its weight acts downwards and the contact force of the stable acts upwards
- If the glass is tilted slightly, the forces are no longer in line



stability

Regular and uniform objects

- The geometrical centre of some common shapes-



- This is also where the centre of gravity and the object's weight, ***W***, can be considered to act

Cont'd

- The Egyptian pyramids are among the wonders of the world. It has been said that if they had been built the other way up, they would have been even greater wonders

Cont’d

Egyptian pyramids

Does the shape of pyramids remind you of a highly stable creation of Allah Ta'ala?

Were you thinking of mountains? Alla'ah Ta'ala states;

(*Have we not made) the mountains as nails (pegs)*

The lower base is very wide as compared to the peak. This is the reason that even a large force cannot turn it round. It is deeply rooted in the ground.

Mountain base is wide

Revision questions

- Explain why double- Decker buses have heavy weights attached to their undersides
- Give a reason why ships fill their holds with ballast if they do not have cargo?
- An alcoholic is not able to balance himself. Support this statement
- Humans can stand upright, is it because of their brain ? Explain your answer

Extension

- Mechanical equilibrium